Roll No. ....

[Total No. of Questions : 05] [Total No. of Printed Pages : 04

ZE-192

B.Tech. (New Scheme) (B.E.)

(Main/ATKT) Examination, 2026

(Fourth Semester)

NUMERICAL ANALYSIS

EL-401

Time : 3 Hours] **[Maximum Marks : 60**

Note : Attempt all questions. All questions carry equal marks. Attempt any two parts from each question.

1. (a) Find a root of $x^3 - 3x - 5 = 0$, correct to three decimal places, using Newton-Raphson method. $x = 2.279018766$

(b) Find the missing term in the following :

x	:	1	2	3	4	5	6	7
y	:	2	4	8	–	32	64	128

(c) Apply Lagrange's formula to find $f(6)$:

x :	2	5	7	10	12
$f(x)$:	18	180	448	1210	2028

2. (a) Find $\int_0^1 \frac{dx}{1+x^2}$ by trapezoidal rule.

(b) Solve the following system of equations by Gauss-elimination method :

$$10x + y + 2z = 13$$
$$3x + 10y + z = 14$$
$$2x + 3y + 10z = 15$$

(c) Solve by Jacobi method :

$$2x - y + 2z = 6$$
$$2x - y + z = 3$$
$$x + 3y - z = 4$$

3. (a) Solve $\frac{dy}{dx} = x + y$, given $y(1) = 0$ to compute y(1.1), using Taylor series method.

(b) Using modified Euler's method, solve for y at $x = 0.1$ from $\frac{dy}{dx} = y^2 + x^2$, $y(0) = 1$.

(c) Using Milne's predictor corrector method find $y(4.4)$, given $5xy' + y^2 - 2 = 0$ and $y(4) = 1$, $y(4.1) = 1.0049$, $y(4.2) = 1.0097$, $y(4.3) = 1.0143$.

4. (a) Find :

(i) $L\{e^{-5t} \cos 3t\}$

(ii) $L^{-1}\left\{\frac{2s+3}{s^2+6s+25}\right\}$

(b) Apply Convolution theorem to find $L^{-1}\left\{\frac{1}{(s^2+a^2)^2}\right\}$.

(c) Explain Fourier transforms and list any two properties of it.

5. (a) The probability density function of a random variable is given by :

$$p(x) = \begin{cases} k & ; \quad x = 0 \\ 2k & ; \quad x = 1 \\ 3k & ; \quad x = 2 \\ 0 & ; \quad \text{otherwise} \end{cases}$$

Find :

(i) k

(ii) $P(X \leq 2)$

(b) Six dice are thrown 729 times. How many times do you expect at least 3 dice to show five or six ?

(c) Find the probability that atmost 5 defective fuses will be found in a box of 200 fuses, if experience shows that 2% of such fuses are defective (given $e^{-4} = 0.0183$).

❖❖❖❖❖❖❖